



FRM NOTE

1. Value at Risk (VaR)

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1. Value at Risk (VaR)

1.1 Portfolio Theory

The traditional mean-variance model estimates the amount of financial risk for portfolios in terms of the portfolio's expected return (i.e., mean) and risk (i.e., standard deviation). Under the mean-variance framework, it is necessary to assume that return distributions for portfolios are elliptical distributions.

One limitation of mean-variance limitation is that the use of the standard deviation as a risk measure is not appropriate for nonnormal distributions. The assumption with financial variables is that they follow a normal distribution. However, for risky financial assets like equities, the distribution tends to have fatter tails and be peaked than the normal distribution. The other probability distribution is positively skewed.

1.2 Value at Risk (VaR)

VaR is interpreted as the worst possible loss under normal conditions over a specific period. Another way to define VaR is an estimate of the maximum loss that can occur with a given confidence level. If an analyst says, "for a given month, the VaR is \$1 million at a 95% level of confidence," then this translates to mean "under normal conditions, in 95% of the months (19 out of 20 months), we expect the fund to either earn a profit or lose no more than \$1 million." By its definition, VaR is ① Estimated return loss ② with a specified confidence level ③ over a specified time period ④ from the frozen portfolio ⑤ under normal market conditions.

Delta-Normal VaR can be computed using the following expression: $[\mu - z\sigma] \times \text{asset value}$. A major limitation of the VaR measure for risk is that two arbitrary parameters are used in the calculation – the confidence level and the holding period. As you can see, the level of risk is dependent on the degree of confidence chosen. VaR increases when the confidence level increases, at an increasing rate.

When the second arbitrary parameter, the holding period, increases the VaR increases. In particular, the rate at which VaR increases is determined in part by the mean(μ) of the distribution. If the return distribution has a $\mu > 0$, then VaR rises at a lower rate and will eventually decrease. If the daily return follows i.i.d normal distribution, T-day return would be $N(T \cdot \mu, T \cdot \sigma^2)$. Then the standard deviation decreases at lower rate because it is squared root of the holding period, T. In this case, this diminishing scale of standard deviation even overestimate risk, when the return has an attribute of mean-reversion.

1.3 Expected Shortfall

Value at Risk is the minimum percent loss, equal to a pre-specified worst case quantile return (typically the 5th percentile return). Expected Shortfall (ES) is the expected loss given that the portfolio return already lies below the pre-specified worst case quantile return. Expected shortfall is also known as conditional VaR (cVaR) or expected tail loss (ETL), and by definition, must exceed VaR.

For a normal distribution with a mean equal to μ and a standard deviation equal to σ , the following equation can be used calculate the expected shortfall:

$$ES = \mu + \sigma \frac{e^{-(z^2/2)}}{(1-x)\sqrt{2\pi}}$$

Unlike VaR, ES has the ability to satisfy the property of subadditivity (하위가법성, $\rho(R_1 + R_2) \leq \rho(R_1) + \rho(R_2)$). Under this property, the risk of a portfolio should never exceed the combined risk of the assets within the portfolio. With VaR, the combined VaR may exceed the summation of the individual assets' VaRs; with expected shortfall, this is not the case.

VaR does not provide any estimate of the magnitude of losses, only the probability that they might occur. When adjusting both the holding period and confidence level at the same time, an ES surface curve showing the interactions of both adjustments is convex.

1.4.1 Linear Derivatives and Nonlinear Derivatives

A linear derivative reflects a relationship between an underlying factor and the derivative that is linear in nature. The delta (rate of change) for a linear derivative must be constant for all levels of the underlying factor, but not necessarily equal to one.

In general, the value at risk (VaR) of a long position in a linear derivative is $VaR_p = \delta \cdot VaR_f$, where VaR_f is the VaR of the underlying factor and the derivative's delta, δ , is the sensitivity of the derivative's price to changes in the underlying factor. How to calculate The delta-normal method explicitly assumes a distribution for the underlying observations which are linearly dependent on the underlying market variables.

Note that there are two categories of risk factors: (1) those where percentage changes are more practical (e.g., stock and commodity prices), and (2) those where actual changes are more practical (e.g., interest rates and credit spreads). Risk factors such as credit spreads or interest rates are more difficult to monitor because they are usually described as a set of points (i.e., a term structure). If certain maturity spreads or interest rates are not observable in the market, linear interpolation can be used to determine the missing rates.

The value of a nonlinear derivative is a function of the change in the value of the underlying asset and is dependent on the state of the underlying asset, and reflects a relationship between an underlying factor and the derivative that is not linear in nature. The change in the value of the call option is dependent in part on how far away the market value of the asset is from the exercise price.

While the delta-normal methods work well for linear portfolios, it is only an approximation for nonlinear products (e.g., call option) or portfolios. The gamma parameters help adjust for the delta's linearity through curvature, therefore delta works much better with in-the-money options than at-the-money options, because the gamma of in-the-money options is much lower: $\Delta P = \delta \Delta S + \frac{1}{2} \gamma (\Delta S)^2$. Contrary to linear products, a normal distribution of the underlying asset price translates into a skewed (nonnormal) distribution for nonlinear products.

Therefore, the delta-normal method (which translates a normal distribution for the asset price into a normal distribution of the derivative) would understate the probability of high option values and would overstate the probability of low option values, which means in both cases investors overestimate the risks.

1.5.1 The Monte Carlo Approach

The Monte Carlo approach generates scenarios using random samples and simulates thousands of valuation outcomes for the underlying assets. Simple example applying for delta-normal approach, a bond's valuation is literally $P_0 = \sum \frac{CF_t}{(1+y)^t}$ and when the yield rate change leads to change the price of the bond, $\Delta P = -P \times Duration \times \Delta y$. Then the VaR of bond price is $VaR(\Delta P) = -P \times Duration \times VaR(\Delta y)$, which makes ΔP the Δy 's function, $f(\Delta y)$, assuming random variable Δy is the only risk factor.

For example, if Monte Carlo produces 500 trials, the daily VaR with a 99% confidence level, will be the fifth worst loss ($= 1\% \times 500$). Let the random variable Δy regressed in stochastic process is like $\Delta y_1 = a(\theta - y) \cdot t + \sigma y_1$, and $y_1 = \epsilon \cdot \sqrt{\Delta t}$, where ϵ is a normal variable chosen by random sampling. We can obtain the result, $VaR(T, X) = VaR(1, X) \times \sqrt{T}$. Monte Carlo simulation can assume any distribution type as long as correlations between the risk factors can be determined.

1.5.2 Stress Testing Methods

In times of crisis, correlations increase (some substantially) and strategies that rely on low correlations fall apart and underestimate risks. A contagion effect often occurs when volatility and correlations both increase, thus mitigating any diversification benefits. One approach for stress testing is to examine historical crisis events. The advantage is that no assumptions of underlying asset returns or normality are needed.

An alternative approach is to analyze different predetermined stressed scenarios. The main advantage of this approach is that it is not limited to the evaluation of risks that have occurred historically. The worst case scenario (WCS) focuses on the distribution of worst possible outcomes given an unfavorable event.

1.6 Parametric vs Nonparametric VaR Methods

VaR is the dollar or percentage loss in portfolio (asset) value that will be equaled or exceeded only x% of time. In other words there are VaR(1%), VaR(5%), VaR(10%), and the risk manager selects the x% probability of interest and the time period over which VaR will be measured. Generally, the time period selected is one day. The VaR(5%) of \$10,000 indicates that there is a 5% chance that on given day, the portfolio will experience a loss of \$10,000 or more. We could also say that there is a 95% chance that on any given day the portfolio will experience either a loss less than \$10,000 or a gain.

The parametric approach requires specific assumptions regarding the asset returns distribution. A parametric model typically assumes asset returns are normally or lognormally distributed with time-varying volatility. The volatility is based on calculating historical variance or standard deviation using “mean squared deviation($E[(r - \mu)^2]$)”. The nonparametric approach is less restrictive in that there are no underlying assumptions of the asset returns distribution. Common nonparametric approaches model volatility using the historical simulation method and the multivariate density estimation method.

2. Market Risk Measures

2.1. Parametric Approach

In contrast to the historical simulation method, the parametric approach (e.g., the delta-normal approach) explicitly assumes a distribution for the underlying observations.

2.1.1 Normal VaR

The VaR cutoff will be in the left tail of the returns distribution. Hence, the calculated value at risk is negative, but is typically reported as positive value since the negative amount is implied.

$$VaR(\alpha\%) = -\mu + \sigma z_\alpha > 0$$

For Example, Assume that the profit/loss distribution for XYZ is normally distributed with an annual mean of \$15 million and a standard deviation of \$ 10 million. VaR at 95%(z=1.65) and 99%(z=2.33) are calculated by a simple calculation. Then we can say that we are 95% confident that the maximum loss will not exceed \$1.5 million.

2.1.2 LogNormal VaR

Suppose the geometric return on Stock follows a normal distribution, then the Stock price would follow the lognormal distribution. The probability that the Stock price is below the VaR is calculated as followed.

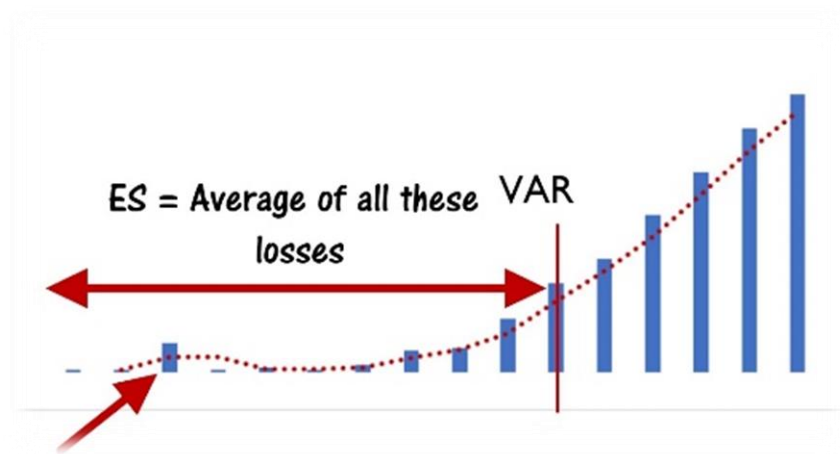
$$P(X \leq VaR) = 1 - e^{-VaR} = 1 - e^{-(\mu + \sigma z)} = 1 - e^{\mu - \sigma z}$$

Then we can calculate lognormal VaR as follows.

$$VaR(\alpha\%) = P \times (1 - e^{\mu - \sigma z})$$

2.2.1 Expected Shortfall (ES)

A major limitation of the VaR measure is that it does not tell the investor the magnitude of the actual loss. The Expected Shortfall (ES) provides an estimate of the tail loss by averaging the VaRs for increasing confidence levels in the tail. For example, the tail mass is divided into n(=5) equal slices and the corresponding n-1(=4) VaRs' average would be the expected shortfall.



Confidence Level	Value at Risk
96%	1.7507
97%	1.8808
98%	2.0537
99%	2.3263
Average (=Expected Shortfall)	2.003

2.3 Coherent Risk Measures

A coherent risk measure(e.g., VaR, ES) is a weighted average of the quantiles of the loss distribution where the weights are user-specific based on individual risk aversion. VaR and ES are equal weighted coherent risk measures. After the quantiles are divided, each quantile is weighted by the specific risk aversion function, as follows.

$$\text{coherent risk measure} = u_1 \text{VaR}_1 + u_2 \text{VaR}_2 + \cdots u_n \text{VaR}_n$$

2.4 Estimating Standard Errors of VaR

It is best practice to also compute the standard error for all coherent risk measures. Is quantile good estimator? First, let's start with a sample size of n and arbitrary bin width of h around quantile, q . Computing standard error is done by realizing that the square root of the variance of the quantile is equal to the standard error of the quantile. After then, a confidence interval for a risk measure can be constructed as follows:

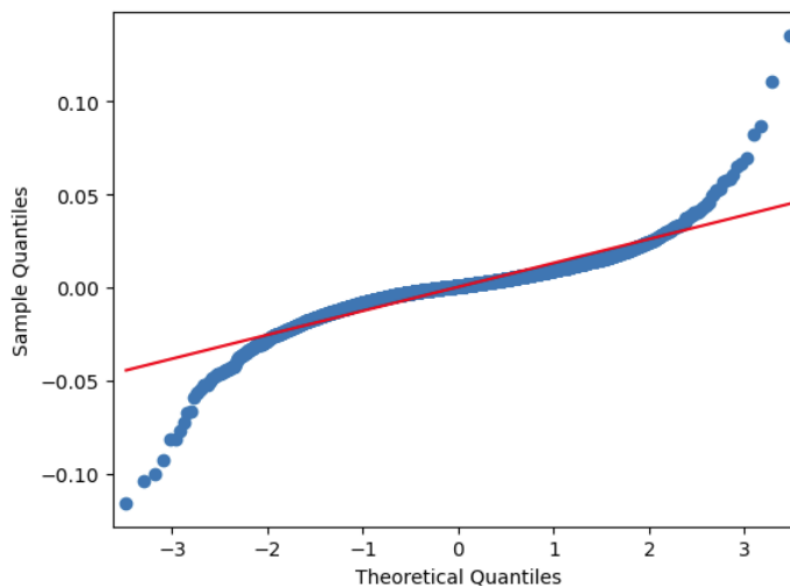
$$[q - se(q)z_\alpha] < \text{risk measure} < [q + se(q)z_\alpha]$$

According to Dowd, 『Market Risk Measurement and Management(2000)』 the standard errors mostly associated with the width of the bin, and the sample size n . It is said that $se(q) = \sqrt{\frac{p(1-p)}{n \cdot f(q)}}$. Suppose we increase the bin size of the histogram, this will increase the frequency of each quantile ($f(q)$) and it leads standard errors ($se(q)$) to decrease, given formula of $se(q)$. And increase of the sample size n also leads $se(q)$ decrease, given formula of $se(q)$, which would makes sense.

Q. Then, what if we control the p , the left tail risk? The book says that $se(q)$ rises when p become more extreme and when we move the tail further.

2.5 Quantile-Quantile (QQ) Plots

In particular, the qq plot is a straightforward way to visually examine if empirical data fits the reference or hypothesized theoretical distribution. The process graphs the quantiles at regular confidence intervals for the empirical distribution against the theoretical distribution.



3. Non-Parametric Approaches

3.1. Bootstrap Historical Simulation

The bootstrap historical simulation is a simple and intuitive estimation procedure. In essence, the bootstrap technique draws a sample from the original data set, records the VaR from that particular sample and “returns” the data. The best VaR estimate from the full data set is the average of all sample VaRs.

3.2. Surrogate Density Function

Historical simulation is unable to incorporate a confidence level of 95.5%, for example. To conquer the discreteness of historical simulation approach, the surrogate density function is used to adjust the histogram. The simplest adjustment is to connect the midpoints between successive histogram bars in the original data set’s distribution. Therefore, the existing data points can be used to “smooth” the data points to allow for VaR calculation at all confidence levels.

3.3 Age-Weighted Historical Simulation

Current VaR may have “ghost effects” of previous events that remain in the computation until they disappear (after n periods), which means that equally weighted historical datasets may not reflect the regime changes occurred in the past. The obvious adjustment to the equal-weighted assumption used in historical simulation is to weight recent observations more and distant observations less. More generally, the weight for an observation that is i days old is equal to:

$$w(1)\{1 + \lambda + \lambda^2 \dots \lambda^n\} = w(1) \cdot \frac{1 \times (1 - \lambda^n)}{1 - \lambda} = 1$$

$$w(i) = \frac{\lambda^{i-1}(1 - \lambda)}{1 - \lambda^n}$$

3.4 Volatility-Weighted Historical Simulation

Another approach is to weight the individual observations by volatility. This was introduced by Hull and White to incorporate changing volatility in risk estimation. The expression is achieved by adjusting each daily return on day t upward or downward based on the then-current volatility forecast, estimated from a GARCH or EWMA model relative to the current volatility forecast on day T . The expression is as followed:

$$return_{T,i} = return_{t,i} \cdot \frac{\sigma_{T,i}}{\sigma_{t,i}}$$

There are several advantages of the volatility-weighted method. First, it explicitly incorporates volatility into the estimation procedure in contrast to other historical methods. Second, the near-term VaR estimates are likely to be more sensible in light of current market conditions. Third, the volatility-adjusted returns allow for VaR estimates that are higher than estimates with the historical data set, which means it reflects the tail risk.

3.5 Correlation-Weighted Historical Simulation

This methodology incorporates updated correlations between asset pairs. When using the historical correlation as a correlation matrix, it needs to be adjusted to the new information environment. The updated variances are utilized in the previous approach as well, but also considering covariances as well as variances.

$$\begin{bmatrix} r_{1t} & r_{2t} \end{bmatrix} \begin{bmatrix} \sigma_{1t} & \sigma_{12t} \\ \sigma_{21t} & \sigma_{2t} \end{bmatrix} \begin{bmatrix} \sigma_{1T} & \sigma_{12T} \\ \sigma_{21T} & \sigma_{2T} \end{bmatrix} = \begin{bmatrix} r_1 & r_2 \end{bmatrix} \begin{bmatrix} \sigma_{1T} & \sigma_{12T} \\ \sigma_{21T} & \sigma_{2T} \end{bmatrix} = \begin{bmatrix} r_{1T} & r_{2T} \end{bmatrix}$$

3.5 Filtered Historical Simulation

The filtered historical simulation is the most comprehensive, and hence the most complicated, of the non-parametric estimators. This process combines the historical simulation model with conditional volatility models (like GARCH or asymmetric GARCH). It is a combination of volatility-weighted historical simulation and bootstrapping method. First, you standardize the returns by volatilities at the time period, calculate VaRs, then keep bootstrapping and average the VaRs.

4. Extreme Value Theory (EVT)

4.1 Generalized Extreme Value (GEV)

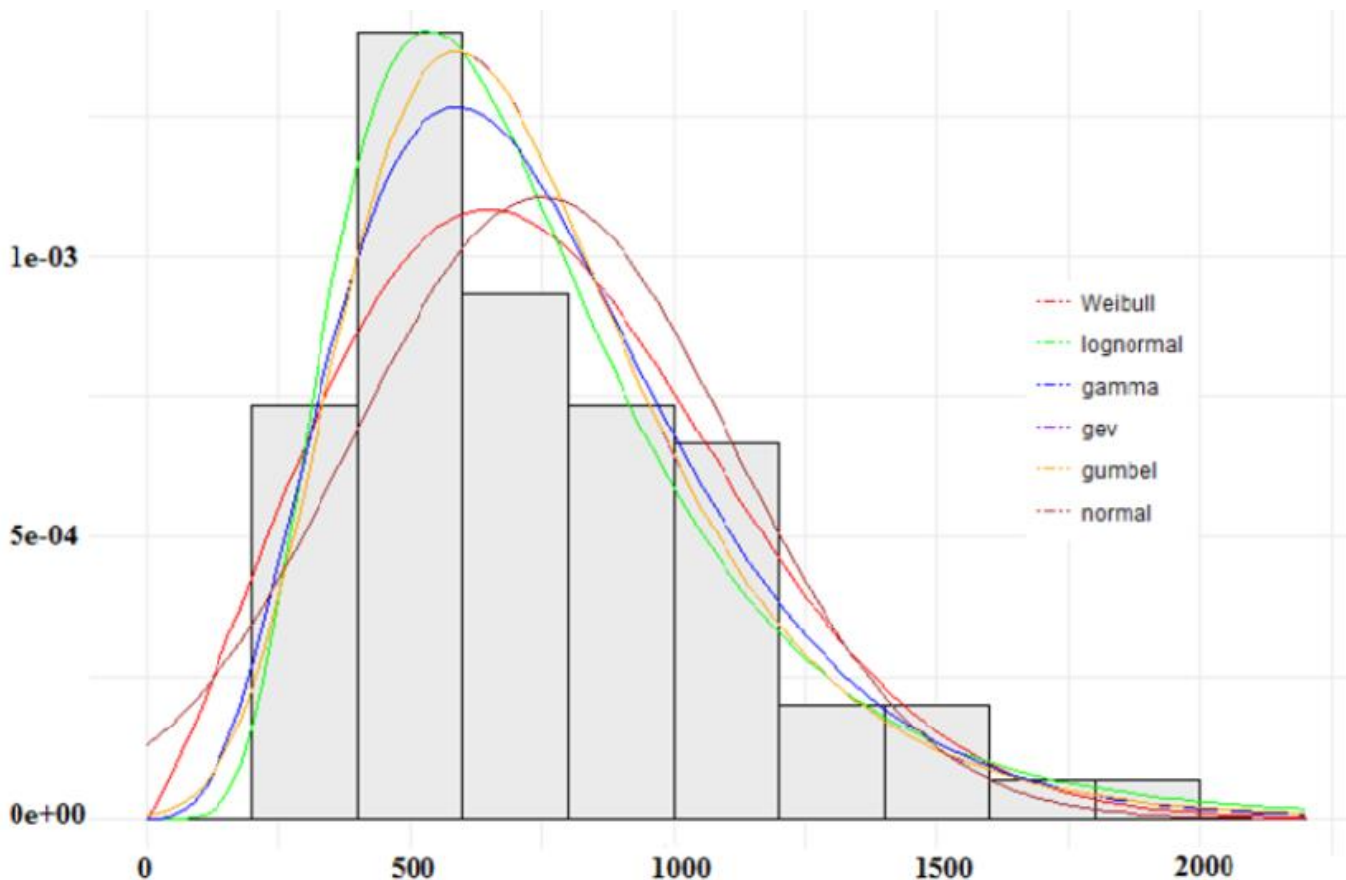
The challenge of analyzing and modeling extreme values is that there are only a few observations for which to build a model, and there are ranges of extreme values that have yet to occur. Researchers usually choose distributions based on measures of central tendency, which misses the issue of trying to incorporate extreme values. Extreme value theory (EVT) focuses on the unique aspects of extreme values and is different from “central tendency” statistics.

One approach for estimating parameters is the Fisher-Trippet theorem. According to the theorem, as the sample size n gets large, the distribution of extremes converges to the following distribution known as the generalized extreme value (GEV) distribution:

$$F(X|\xi, \mu, \sigma) = \exp \left[-1 \left(1 + \xi \left(\frac{x-\mu}{\sigma} \right)^{-\frac{1}{\xi}} \right) \right] \text{ if } \xi \neq 0,$$

$$F(X|\xi, \mu, \sigma) = \exp \left[-\xi \left(\frac{x-\mu}{\sigma} \right) \right] \text{ if } \xi = 0,$$

$$\text{where } 1 + \xi \left(\frac{x-\mu}{\sigma} \right) > 0$$



if $\xi > 0$, the GEV becomes a Frechet distribution, and tails are “heavy” as is the case for the t-distribution and Pareto distributions. if $\xi = 0$, the GEV becomes a Gumbel distribution, and tails are “light” as is the case for the normal and lognormal distributions. if $\xi < 0$, the GEV becomes the Weibull distribution, which is not the case in reality. It contains 3 EVT parameters, ξ (shape parameter), μ (location parameter) and σ (scale parameter). EVT parameters are actually abstracted through regression analysis, not by algebra.

4.2 Peaks-Over-Threshold

The peaks-over-threshold (POT) approach is an application of extreme value theory to the distribution of excess losses over a high threshold. The block maxima is a method used to analyze and model extreme events by focusing on the largest observations within fixed-size blocks of time or space. Block maxima in time series analysis is to find extreme values in an equally divided time horizons, but the block maxima in POT is to find extreme values that are greater than a threshold.

The POT approach generally requires fewer parameters, ξ (shape parameter), ν (scale parameter). The point is how to select the adequate threshold. If the threshold is too high, the model verification is not applicable because not too much extreme values are explored and it needs more sufficient dataset. If the threshold is too low, too much extreme values are explored.

```
# Peaks-Over-Threshold

import numpy as np

np.random.seed(42)
data = np.random.normal(loc=0, scale=1, size=1000)

threshold = 1.5

ev = data[data > threshold]

print(ev)
```

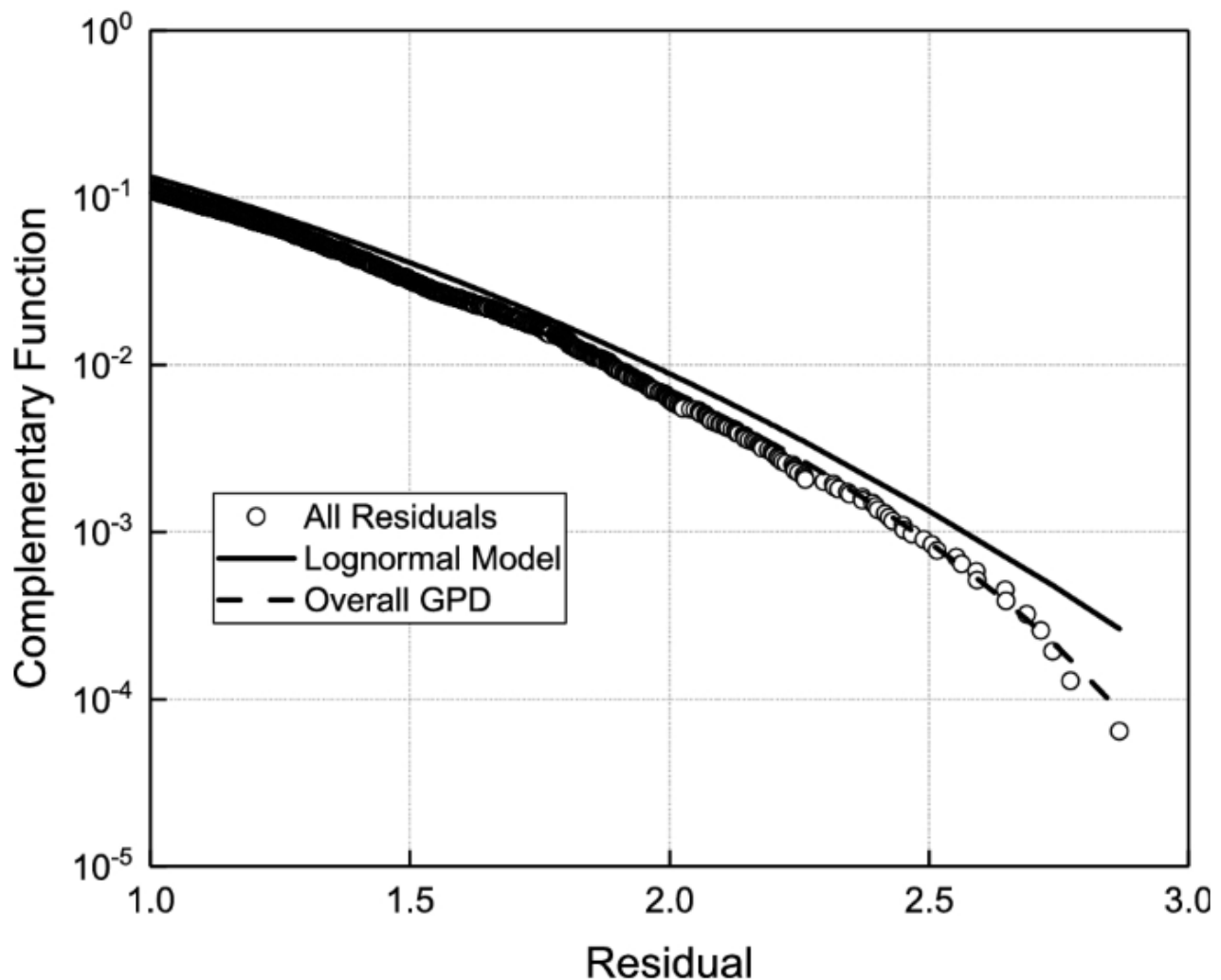
4.3 Generalized Pareto Distribution

The Gnedenko-Pickands-Balkema-deHaan (GPBdH) theorem says that as u gets large, the distribution converges to a generalized Pareto (GP) distribution, such that:

$$1 - [1 + \frac{\xi x}{\beta}] \text{ if } \xi \neq 0,$$

$$1 - \exp [-\frac{x}{\beta}] \text{ if } \xi = 0,$$

The GP distribution exhibits a curve that dips below the normal distribution prior to tail. Since all distributions of excess losses converge to the GP distribution, it is the natural model for excess losses, which reflects that a few extreme value determines the parent distribution. The GP distribution requires 2 parameters, ξ (shape parameter), β (scale parameter).



Multivariate EVT

Multivariate EVT is important because we can easily see how extreme values can be dependent on each other. Multivariate EVT has the same goal as univariate EVT in that the objective is to move from the familiar central-value distributions to methods that estimate extreme events. Assumptions of an elliptical distribution and the use of a covariance matrix are of limited use for the multivariate EVT.

Modeling multivariate extremes requires the use of copulas. Multivariate EVT says that the limiting distribution of multivariate extreme values will be a member of the family of EV copulas, and we can model multivariate EV dependence by assuming one of these EV copulas.



5. VaR Mapping

VaR mapping involves replacing the current values of a portfolio with risk factor exposures. First measure all current positions within a portfolio, these positions are then mapped to risk factors by means of factor exposures. By reducing the number of variables under consideration, we greatly simplify the risk management process.

So, how many general risk factors (or primitive risk factors) are appropriate for a given portfolio? Of course, the more risk factors chosen, the more time consuming the modeling of a portfolio becomes. Specific risks arise from unsystematic risk or asset-specific risks of various positions in the portfolio. The more precisely we define risk, the smaller the specific risk. Thus, the definition of specific risk is a function of general market risk.

After we have selected our general risk factors, we must map our portfolio onto these factors. The three methods of mapping for fixed-income securities are (1) principal mapping, (2) duration mapping, and (3) cash flow mapping.

5.1 Principal mapping

Principal mapping is the simplest technique as it only considers the timing of the redemption or maturity payments of the bonds. While it simplifies the process, it ignores all coupon payments of the bonds. Principle mapping only considers the average life of the portfolio.

Maturity	VaR %
1	0.4696
2	0.9868
3	1.4841
4	1.9714
5	2.4261

Suppose a portfolio consists of two par value bonds. One bond is a one-year \$100 million bond with a coupon rate of 3.5%. The second bond is a five-year \$100 million bond with a coupon rate of 5%. The risk percentages (or VaR percentages) for zero-coupon bonds with maturities one to five years are as follows.

The weights in this example are both 50% (i.e., \$100 million / \$200 million). Thus, the weighted average life of this portfolio for the two bonds is three years $[0.5(1)+0.5(5) = 3]$. Assuming the total portfolio value of \$200 million, The VaR percentage at the 95% is 1.4841 for a three-year zero-coupon bond. VaR computed by the principal mapping is as follows:

$$\text{Principal mapping VaR} = \$200 \text{ million} \times 1.4841\% = \$2.968 \text{ million}$$

5.2 Duration Mapping

To calculate the VaR of this fixed-income portfolio using duration mapping, we simply replace the portfolio with a zero-coupon bond that has the same maturity as the duration of the portfolio.

Year	CF of 1y bond	CF of 5y bond	Spot rate	PV(CF)	$t \times \text{CF}$
1	\$103.5	\$5	3.50%	\$104.83	\$104.83
2	\$0	\$5	3.90%	\$4.63	\$9.26
3	\$0	\$5	4.19%	\$4.42	\$13.26
4	\$0	\$5	4.21%	\$4.24	\$16.96
5	\$0	\$105	5.10%	\$81.88	\$409.38

The table above demonstrates the calculation of Macaulay duration for this portfolio. Duration is then computed as \$553.69 million / \$200 million = 2.768. The next step is to interpolate the VaR for a zero-coupon bond with a maturity of 2.768 years. The VaR of a 2.768 maturity is interpolated as follows: $0.9868\% \times (1.4841\% - 0.9868\%) \times (2.768 - 2) = 1.3687\%$. Substituting the VaR percentage, we can calculate the Duration mapping VaR.

$$\text{Duration mapping VaR} = \$200 \text{ million} \times 1.3687 = \$2.737 \text{ million}$$

5.3 Cash Flow Mapping

In order to calculate the VaR for this fixed income portfolio using cash flow mapping, we need to map the present value of the cash flows (i.e., face amount discounted at the spot rate for a given maturity) onto the risk factors for zeros of the same maturities and include the inter-maturity correlations.

If the five zero-coupon bonds are all perfectly correlated ($\rho = 1$), then the undiversified VaR could be calculated as follows:

$$\text{Undiversified VaR} = \sum_{i=1}^N |x_i| \times V_i$$

If the five zero-coupon bonds are not all perfectly correlated ($\rho \neq 1$), the diversified VaR using correlation matrix is calculated as below, where x is the present value of cash flows vector, V is the vector of VaR for zero-coupon bond returns and R is the correlation matrix.

$$\text{Diversified VaR} = \sqrt{(x \times V)^T R (x \times V)}$$

Year	x	$x \times V$	1Y	2Y	3Y	4Y	5Y	$x\Delta\text{VaR}$
1	104.83	0.4923	1	0.894	0.887	0.871	0.861	1.17
2	4.63	0.0457	0.894	1	0.99	0.964	0.954	0.115
3	4.42	0.0656	0.887	0.99	1	0.992	0.987	0.170
4	4.24	0.0836	0.871	0.964	0.992	1	0.996	0.217
5	81.88	1.9864	0.861	0.954	0.987	0.996	1	5.168

In this example, the undiversified VaR is computed as the sum of the third column: 2.674. The diversified VaR is the square root of the sum of the last column (6.840) and is calculated as 2.615. If we assume that there is perfect correlation among maturities of the zeros, the portfolio VaR would be equal to the undiversified VaR (i.e., the sum of the VaRs). Stress testing each zero by its VaR is a simpler approach than incorporating correlations; however, this method ceased to be viable if correlations are anything but perfect.

5.4 Tracking Error VaR

It is often convenient to measure VaR relative to a benchmark portfolio. The VaR of the deviation between the two portfolios is referred to as a tracking error VaR. In other words, tracking error VaR is a measure of the difference between the VaR of the target portfolio and the benchmark portfolio. While the absolute VaR is computed by multiplying the VaR percentage times the market value of price of assets in the portfolio, we define the new vector of market value positions for each portfolio as x and the vector of market value positions of the benchmark as x_0 . Then the relative performance to the benchmark is computed as tracking error VaR as follows:

$$\text{Tracking error VaR} = \alpha \sqrt{(x - x_0)' \Sigma (x - x_0)}$$

Though the absolute VaR is the smallest among the portfolios, that does not imply that the portfolio's tracking error VaR is the smallest since portfolio VaR has a tendency to shrink when the portfolio's cash flow is closely aligned with the benchmark portfolio.

Tracking error VaR can be used to compute the variance reduction (similar to R-squared in a regression) as follows:

$$\text{Variance improvement} = 1 - \left(\frac{\text{Tracking error VaR}}{\text{Benchmark VaR}} \right)^2$$

5.5 Forward Contracts

Forwards are appropriate for the application of the delta-normal method. Their values are a linear combination of a few general risk factors, which have commonly available volatility and correlation data. Suppose you purchase 100 million euros and sell \$126.5 million USD and spot rate $S_{USD/EUR}$ is 1.25, with forward rate $F_{USD/EUR}$ being 1.265. For euro investor, the forward contract is a synthetic position of 3 risk factors, which are long EUR spot, long EUR bill and short USD bill.

Risk Factor	Price/Rate	VaR%	EUR Spot	1y EUR	1y USD
EUR Spot	1.2500	4.5381	1.000	0.115	0.073
Long EUR bill	0.0170	0.1396	0.115	1.000	-0.047
Short USD bill	0.0292	0.2121	0.073	-0.047	1.000

In this example, we have a long position in a EUR contract worth \$122.91 million (=100 million euros / 1.0170) and a short position in a one-year U.S. T-bill worth \$122.91 million (=\$126.5 / 1.0292) today.

Position	PV factor	CF	X	$ x_i V_i$	$x\Delta VaR$
EUR spot			122.911	5.578	31.116
Long EUR bill	0.9777	100.0	122.911	0.172	0.142
Short USD bill	0.9678	126.5	122.911	-0.261	-0.036

The undiversified VaR for this position is \$6.01 million (=5.578+0.172+|-0.261|), and the diversified VaR for this position is \$5.588 million ($=\sqrt{31.116 + 0.142 - 0.036}$).

5.6 Forward Rate Agreement (FRA)

Suppose you have an FRA that locks in an interest rate one year from now, selling 6×12 FRA on \$100 million. Assuming that the 360-day spot rate is 4.5% and the 180-day spot rate is 4.1%, the present values of the notional \$100 million contract is $\$100 \text{ million} / 1.0205 = \97.991 million . This will be invested for a 12-month period. The forward rate is computed as $f_{1,2} = 1.045 / (1 + 0.041 / 2) = 4.8\%$.

Position	PV(CF), x	VaR%	180d	360d	$ x_i V_i$	$x\Delta VaR$
180 days (borrow)	-97.991	0.1629	1	0.79	-0.160	-0.0325
360 days (invest)	97.991	0.4696	0.79	1	0.460	0.1537

The sixth column computes the undiversified VaR of \$0.62 million at the 95% confidence level using VaR percentages in the third column multiplied by the absolute value of the present values of cash flows. Matrix algebra is then used to multiply this vector by the correlation matrix presented in columns four and five to compute the diversified VaR of \$0.348 ($=\sqrt{-0.0325 + 0.1537}$).

5.7 Interest Rate Swaps (IRS)

Interest rate swaps are commonly used to exchange interest rates from fixed to floating rates or from floating rates. Suppose you want to compute the VaR of a \$100 million four-year swap that pays a fixed rate for four years in exchange for a floating rate-payment. This vanilla swap's payer, who pay fixed and receive floating, has two positions: short fixed bond and long FRN. Just following the same procedures above, the undiversified VaR is sum of the VaRs of fixed and floating and the diversified VaR is the squared root of VaRs and covariance matrix between the fixed and float.

5.8 Nonlinear Derivatives

Options exhibit nonlinear relationships between movements of the values of the underlying instruments and the values of options. In many cases, the delta-normal method may still be applied because the value of an option may be expressed linearly as the product of the option delta and the underlying asset. Over the short periods of time, such as one day, a linear approximation of the delta is more accurate. For example, assume the strike price of an option is \$100 with a volatility of 25%. If we are only concerned about a one-day horizon, then the one-day loss could be computed as follows:

$$\text{option 1d VaR} = -\alpha \cdot S \cdot \sigma \sqrt{T} = -1.645 \times \$100 \times 0.25 \left(\sqrt{\frac{1}{252}} \right) = -\$2.59$$

6. Backtesting VaR

Backtesting is the process of comparing losses predicted by a value at risk (VaR) model to those actually experienced over the testing period. It is an important tool for providing model validation, which is a process for determining whether a VaR model is adequate. The number of actual observations that fall outside a given confidence level are called exceptions. The number of exceptions falling outside of the VaR confidence level should not exceed one minus confidence level.

VaR models are based on static portfolios, while actual portfolio compositions are constantly changing as relative prices change and positions are bought and sold. Another difficulty with backtesting is that the sample backtested may not be representative of the true underlying risk.

The VaR modeled returns are comparable to the hypothetical return that would be experienced had the portfolio remained constant for the holding period. Generally, we compare the VaR model returns to cleaned returns (i.e., actual returns adjusted for all changes that arise from changes that are not marked to market, like funding costs and fee income).

6.1 Failure Rates

If a VaR model were completely accurate, we would expect VaR loss limits to be exceeded (this is called exception) with the same frequency predicted by the confidence level used in the VaR model. Thus, backtesting is the process of systematically comparing actual (exceptions) and predicted (model) loss levels.

Failure rates define the percentage of times the VaR confidence level is exceeded in a given sample. An unbiased measure of the number of exceptions as a proportion of the number of samples is called the failure rate. The probability of exception, p , equals one minus the confidence level ($p = 1 - c$). If we use N to represent the number of exceptions and T to represent the sample size, the failure rate is computed as N / T . When T (the sample size) gets larger, the failure rate converges to the significance level, p .

Testing that the model is correctly calibrated requires the calculation of a z-score, where $z = \frac{x - pT}{\sqrt{p(1-p)T}}$ and then z-score is compared to the critical value at the chosen level of confidence. Suppose daily revenue fell below a predetermined VaR level (at the 95% confidence level) on 22 days during a 252-day period. Is this sample an unbiased sample? The answer is no (in precise, we would reject the null hypothesis that the VaR model is biased).

$$z = \frac{22 - 0.05(252)}{\sqrt{0.05(0.95)252}} = 2.72 > 1.96$$

6.2 Type I and Type II Errors

When determining a range for the number of exceptions that we would accept, we must strike a balance between the chances of rejecting an accurate model (Type I error) and the chances of failing to reject an inaccurate model (Type II error). The goal of backtesting is to create a VaR model with a low Type I error and include a test for a very low Type II error, which means that it values Type II error more.

T=250	Coverage = 99%		Coverage = 97%	
	Model is Correct		Model is Incorrect	
N (# of exceptions)	PDF (X=N)	CDF (X≥N)	PDF (X=N)	CDF (X<N)
0	8.1059	100	0.0493	0
1	20.4693	91.8941	0.3813	0.0493
2	25.7417	71.4248	1.4681	0.4306
3	21.4948	45.6831	3.7534	1.8987
4	13.4071	24.1883	7.1682	5.6521
5	6.6629	10.7812	10.9074	12.8203
6	2.7482	4.1183	13.7749	23.7277
7	0.9676	1.3701	14.8501	37.5026
8	0.2969	0.4025	13.9507	52.3527
9	0.0806	0.1056	11.6016	66.3034
10	0.0196	0.025	8.6474	77.905
11	0.0043	0.0054	5.8351	86.5524
12	0.0009	0.0011	3.5943	92.3875

THE TABLE IS ABOUT THE BASEL RULES FOR BACKTESTING, PROBABILITIES OF OBTAINING EXCEPTIONS (T = 250). THE PDF IS CALCULATED BY USING BINOMDIST FUNCTION IN EXCEL. THE BASEL COMMITTEE ALLOWS THE NUMBER OF EXCEPTIONS LESS THAN 5, WHICH ALSO MEANS THAT THE TYPE I ERROR IS UNDER 10.7812 AND TYPE II ERROR UNDER 12.8203. THE BANKS HAVE AN INCENTIVE TO LOWER THE CAPITAL REQUIREMENT, WHICH IS RELATED TO TYPE II ERROR.

6.3 Unconditional Coverage

Kupiec (1995) determined a measure to accept or reject models using the tail point of a log-likelihood ratio (LR) as follows:

$$\text{It is known that } 1 < \frac{P(\frac{N}{T})}{P(p)} = \frac{(\frac{N}{T})^N \cdot (1 - \frac{N}{T})^{T-N}}{p^N \cdot (1 - p)^{T-N}}$$

Take log and square for both sides,

$$0 < LR_{UC} = -2 \ln[(1 - p)^{T-N} p^N] + 2 \ln \left\{ \left[1 - \frac{N}{T} \right]^{T-N} \left[\frac{N}{T} \right]^N \right\} \sim \chi_1^2$$

where p is the probability level, T is the sample size, N is the number of exceptions, and LR_{UC} is the test statistic for unconditional coverage (uc). The term unconditional coverage refers to the fact that we are not concerned about independence of exception observations or the timing of when the exceptions occur. We simply are concerned with the number of total exceptions. We would reject the hypothesis that the model is correct if the $LR_{UC} > 3.84 (= 1.96 \times 1.96)$.

The unconditional coverage illustrates how increasing the sample size allow us to reject the model easily, and how difficult it is to backtest VaR models constructed with higher level of confidence. For example the non-rejection region of the sample size with 252 is $N < 7$, while sample size 1000 is $4 < N < 17$. And the non-rejection region at the significance level 0.01 is $N < 7$, while at significance level 0.025 it is $2 < N < 12$.

6.4 Conditional Coverage

Conditional Coverage is the measure when including the measure of the independence of exceptions. Christofferson proposed extending the unconditional coverage test statistic to allow for potential time variation of the data. The overall log-likelihood test statistic for conditional coverage is then computed as:

$$LR_{CC} = LR_{UC} + LR_{IND} \sim \chi_2^2$$

Each individual component is independently distributed as chi-squared, and the sum is also distributed as chi-squared. At the 95% confidence level, we would reject the model if $LR_{CC} > 5.99$. And we can also reject the independence term alone if $LR_{IND} > 3.84$, which means that the exceptions are serially dependent.

6.5 Basel Committee Rules

The Basel Committee is primarily concerned with identifying whether exceptions are the result of bad luck (Type I error) or a faulty model (Type II error). The Basel Committee requires that market VaR be calculated at the 99% confidence level and backtested over the past year. To mitigate the risk that banks willingly commit Type II error and use a faulty model, the Basel Committee designed the Basel penalty zones, which apply penalty if the number of exceptions are more than 4. However, industry analysts have suggested lowering the required VaR confidence level to 95% and compensating by using a greater multiplier.